Computational genomics: HW 5 – Final Project (50%)

Background

Prostate cancer is cancer that occurs in the prostate. The prostate is a small walnut-shaped gland in males that produces the seminal fluid that nourishes and transports sperm.
Prostate cancer is one of the most common types of cancer. Many prostate cancers grow slowly and are confined to the prostate gland, where they may not cause serious harm. However, while some types of prostate cancer grow slowly and may need minimal or even no treatment, other types are aggressive and can spread quickly.
Prostate cancer that's detected early — when it's still confined to the prostate gland — has the best chance for successful treatment.

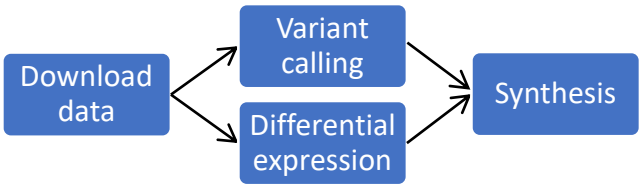
[Mayo Clinic]

To strengthen the public image of the Israeli science community – a task of great importance these days, no doubt – the government has sent you to China to underlie the mechanism behind prostate cancer in the Chinese population.

You will analyze 14 primary prostate cancers and their paired normal counterparts from the Chinese population using RNA-seq data to achieve your goal.

A schematic overview of the assignment

A. Download data from the SRA and process it as needed.
B. Separately call for variants in normal tissues and tumor tissues, and by comparing tumor variants to normal variants and known variants (CleanVar SNPs), retain only novel, somatic variants presumably caused by prostate cancer.
C. Analyze the differential expression between the groups and test for enrichment in significantly changed genes.
D. Finally, using only positively significantly changed genes (i.e., whose expression levels are significantly higher), count the number of novel, somatic variants (from step B) per each such gene. Pick one gene with many variants and explain the possible connection between increased expression levels and the creation of somatic mutations in that gene.



Required outputs

1. A supplementary PDF file containing images and screenshots as necessary.
2. A markdown file combining detailed textual explanations and code snippets of every step you did. By detailed explanations, we mean that you need to reason about each and every step. Also, refer to images in the supplementary PDF file when required.

Note: you need to submit both files.

General notes

1. Personal integrity & submission guidelines:
    a. No cheating. Thanks.
    b. You may use ChatGPT just as you can use Google, but it's your sole responsibility to verify that the code it suggests works!
    c. Also, whether you use Chat GPT or are highly verbose, provide us with just enough explanations to prove you understand what you did. These explanations need to be written in your own words.
    d. Except for downloading and uploading the tables from your PC to the bastion host, you must use code (bash/python/R) for every other step.
    e. Take extra care to treat the markdown file as your main output file (i.e., combine code, results, and explanations for why you used the code as you did according to the results). Also, properly format the markdown file (titles, code inside code snippets, and text outside code snippets…).

    Points will be taken down for **any** violation of these requirements.
2. We use the term VCF interchangeably for VCF and BCF files.

Tips

1. To save on disk space:
    a. Compress FastQ files using *gzip* and convert SAMs to BAMs.
    b. Respectively delete intermediate FastQ and SAM files.
2. To save on machine hours, try to utilize all your CPUs, whether by increasing the number of used threads or running in parallel using *xargs* (demonstrated in class).

Input files

In this assignment, you will work with the following files/tables:

	File name	What's in the file?	Previously seen at
1	SraRunTable.HW5.Fixed.txt	An SRA run table defining the samples and their conditions	
2	Genes.RefSeq.JAN24.bed.gz	Genes bed file	HW3
3	SNPs.ClinVar.JAN24.bed.gz	SNPs bed file	HW3

4	GeneIDGeneSymbol.txt	A table mapping between gene symbol and gene ID	HW3
5	TranscriptToGene.hg38.tsv	A table mapping between gene symbol and transcript	Practices 8-9
6	transcript_to_gene.py	A script to sum per-gene read counts and TPM levels from multiple Salmon outputs	Practices 8-9

Steps

1. Using *fastq-dump*, download to your class machine 3,000,000 reads from each RNA-seq sample described in the SRA run table (file 1).

2. Contrary to the previous one, cleaning the data is a must in this assignment. It's only natural – the data is over 10 years old and considered **ancient** in computational biology time scales. So, the question is not whether to clean it but how to do it. Take care to consider the experimental design (14 coupled samples, a normal tissue, and a tumor tissue from each of the 14 participants) when doing so.
   (Hint: due to the large number of samples, removing a few very low-quality samples may be beneficial.)

   Therefore, investigate the raw data quality using *fastqc* and *MultiQC*, and use *fastp* to clean it the way you think is necessary, and verify (again, using *fastqc* and *MultiQC*) that the data's quality is now to your liking.
   Please explain your decisions regarding the parameters chosen in specific samples and supplement them with images from the *MultiQC* report(s).

3. Use *bwa mem* to align the data to the human genome and save the alignment in BAM format, sorted, and indexed.
   (Admittedly, recall that *bwa mem* is not the best tool for the job, but for the sake of simplicity, it will suffice.)
   (It is recommended that you use *samtools flagstat* to ensure you have achieved a successful alignment.)

4. Use *bcftools* to call variants in each sample and then merge the individual VCF files by groups, that is, one merged file for normal tissues and one for tumor ones.
   Use only high-quality variants (e.g., with good quality scores and sufficient coverage). Explain the chosen parameters.

5. Now, we would like to retain only somatic and novel SNP variants. That is, only SNPs that appear in tumor tissues but not in normal ones ("somatic") and that weren't identified before ("novel").

   We make the following assumptions:

(a) Consider a position X at chromosome Y in any merged VCF file. If any alternative alleles in this position are SNPs, we consider the position a **tissue** variant.
(b) We only compare SNPs by their coordinates, neglecting mismatch types.

Therefore, by retaining only positions with SNPs from each tissue's VCF, we can compare SNPs between the two tissues and known SNPs (file 3).
(Hint 1: use *bcftools view* with *--types snps* to satisfy assumption (a).)
(Hint 2: use *bedtools* to intersect between the VCFs and the known SNPs bed file.)
Save that VCF file to the disk.

6. Use *Salmon* to quantify per-sample expression levels. Then, use the script *transcript_to_gene.py* to obtain one file containing the per-sample per-gene reads counts and TPM levels.

7. Use *PyDESeq2* to analyze differential expression between the tissues, using cutoffs of $\left|log_2\left(foldchange(x)\right)\right| \geq 2$ and $-log_{10}\left(padj(x)\right) \geq -log_{10}(5 * 10^{-5})$, where x is a gene.
   Create two gene lists and save them to the disk:
   (a) Significantly changed genes (according to cutoffs).
   (b) Significantly **positively** changed genes, that it, genes whose expression levels increased in tumor tissues significantly.

8. Supply us with the following plots:
   (a) A volcano plot of all the genes, with lines drawn according to the cutoffs above.
   (b) **BONUS**: A clustergram of the significantly changed genes (again, according to cutoffs). A success will award you a 10-point increment; however, the final score won't exceed 100 points.
   (c) Use the *metascape* website to perform GO analysis on the significantly changed genes, and choose the first plot in that output, which is a bar plot of enriched GO terms across the input genes list, colored by p-values.

9. Finally, we want to combine our insights from novel somatic variants and differentially expressed genes in prostate cancer.
   Using only positively significantly changed genes (from step 7), count the number of novel, somatic variants (from step 5) per each such gene.
   Please supply us with a list of the 20 genes with the most such SNPs inside, sorted by descending order of the number of SNPs.

   Do this as follows :
   (a) Link the gene symbols of the positively significantly changed genes to their gene ID equivalents (file 4) and use that to link these genes to their coordinates in the genes bed

file (file 2).
(b) Sort the gene coordinates from (a) and merge them by running `bedtools merge -i <path_to_file> -s -c 4,5,6 -o distinct, sum,distinct`. Observe the output and explain what that merge command did.
(c) Intersect the merged gene coordinates from (b) with the novel somatic variants VCF file and count the number of SNPs per interval.
(d) If you did everything right, the output of (c) is the sum of SNPs per overlapping gene IDs. In each line, remap the gene IDs from their gene symbol. Ideally, whenever gene IDs' coordinates overlap, they are all isoforms of the same gene, marked by a gene symbol. However, that's not always the case, so you need to remove such lines.

In addition, pick one gene with many variants and explain the possible connection between increased expression levels and the creation of somatic mutations in that gene. In your answer, rely on scientific literature and cite/supply links accordingly.

The answer should include text and screenshots supporting your claims.

Good luck!